



ATI Catalyst Control Centre And NVIDIA nTune

Unleash hell with your red monster or turbo-charge your green devil

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A good gamer needs to extract the maximum performance from his gaming hardware, the most important one being the graphics card. Today ATI and NVIDIA are the two leading graphics card manufacturers in the world, and tools that will let you overclock and optimise graphics cards from them are easily available. There are far fewer tools available to tweak ATI hardware, but this is because ATI's drivers, called Catalyst, provide more than enough options to tweak your graphics cards till they sweat. To cater to the enthusiasts, NVIDIA has released its own set of tools as well—nTune—to tweak its motherboards and graphics cards. Obviously, we prefer using tools that are made by the manufacturers, because they allow you to tune and tweak the cards to the limits, and do so safely.

In this article, we'll look at some important settings for ATI Catalyst and NVIDIA nTune, to help you quickly understand and use these software. We'll begin with ATI.



ATI Catalyst

We assume that you are using an ATI graphics card which is no older than the RADEON. Though you can use any recent Catalyst or modified driver, we are going to stick to the latest official 7.9 version. We will attempt to cover all the important aspects of the Catalyst Control Panel (henceforth referred to as CCC). You can get the latest ATI Catalyst drivers at <http://ati.amd.com/support/driver.html>. Opt to download the Catalyst drivers along with their Catalyst Control Center (CCC). You may also need to install the Microsoft .NET Framework 1.1 (<http://tinyurl.com/73qzr>).

Catalyst Control Panel

To open the CCC, right-click on the system tray icon and click on Catalyst Control Panel. When you open the CCC for the first time, you will be asked whether you want to use the Basic or Advanced view. We suggest that you



The ATI Catalyst Control Center

choose Advanced view since it enables greater control and also check the box "Don't show this page again" and click Next.

Profiles

The Profiles button launches the Profile Manager, where you can save your current set of preferences under a profile name. This allows



Display Options of CCC

you to use different graphics card settings in different circumstances. Click on the Profiles button, select Profiles Manager and save your profile using any name.

GRAPHICS SETTINGS

Now let us examine the Graphics Settings, the meat of the CCC.

Display Options

Under Display Options, you will find the 3D Refresh Rate Override option that allows you to force a game's refresh rate to exceed 60 Hz. This can

be done by selecting "Same as desktop", and is especially useful if you use a CRT monitor.

Monitor Properties

Next we go on to Monitor Properties. The first sub-section is Attributes, where you can choose to use "Use Extended Display Identification Data (EDID)", which is recommended if available. This option allows the display device to provide accurate information about itself to the Catalyst drivers and will therefore save you from setting inappropriate values for resolutions and refresh rates, saving your monitor from damage. The second sub-section is Adjustments, where you can adjust the position of the screen, vertically as well as horizontally and also its height and width. These controls are useful to CRT monitor users. If you are sure that your monitor supports HDTV resolutions, then you can unlock these in the HDTV Support sub-section. After unlocking here, you can add these resolutions using the Force button in the Displays Manager.

Digital Panel Properties

If you have an LCD monitor connected to the graphics card, the Digital Panel Properties section will be visible. You can view the various details about your LCD monitor such as panel name, link type, native resolution and the

refresh rate in the Attributes sub-section. In Image Scaling section, you can choose Scale to full panel size to scale full-screen applications to occupy the complete screen even if the application resolution is not the same as the native resolution of the LCD. This may sometimes produce images that appear a little blurred though. Choose Use centred timings in such cases, which will display the application screen at the actual resolution. Thus if an application has a resolution lower than that of the LCD, it will appear as a box at the centre of the screen surrounded by black borders, and if it is larger, it will cause the screen to scroll. Lastly, there is the DVI Settings section, where you can select Reduce DVI frequency on high-resolution displays and Alternate DVI operational mode settings to reduce or remove any potential graphical anomalies when you are using a resolution above 1280x1024. Most panels may not need these options and these can be left un-ticked by default.



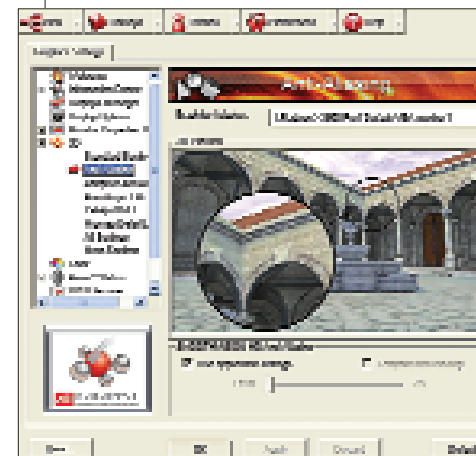
Digital Panel Properties can be adjusted as needed

ational mode settings to reduce or remove any potential graphical anomalies when you are using a resolution above 1280x1024. Most panels may not need these options and these can be left un-ticked by default.

3D SETTINGS

Anti-aliasing

There are many sub-sections in 3D settings, the first one being Standard Settings, where you can tick the checkbox next to "Use custom settings" and move a slider to choose from amongst five steps ranging from Performance to Quality. A preview window shows exactly how the setting that you choose will affect the image quality and this is present in each of the



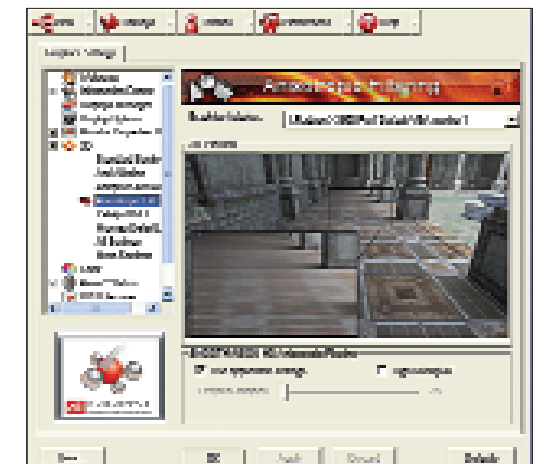
Tweak Anti-aliasing settings for smoother edges

settings in this section. This setting is for those who are not comfortable fiddling with the advanced graphics settings. Speaking of advanced settings, which begin next, we go on to Anti-Aliasing. Anti-aliasing determines how jagged the edges of an object will appear. Uncheck the Use application setting checkbox and move the slider to the right for a smoother edges. Crossfire users will see more options if this feature is being used. Enabling Temporal anti-aliasing gives the effect of a higher degree of anti-aliasing but with a lower performance hit (unlike the

normal anti-aliasing). Thus, a 2x temporal anti-aliasing will look almost like 4x normal anti-aliasing when the game runs at high frame rates. When you enable adaptive anti-aliasing, the visual quality of scenes containing transparent textures will improve. Here you can choose between Quality and Performance modes.

Anisotropic Filtering

Anisotropic filtering is a technique that sharpens textures as they fade into the distance. You must uncheck the "Use application settings" checkbox to use it and move the slider to the right to increase the filtering. You can enable High Quality AF to enable trilinear instead of bilinear filtering, but this will have minimal effect on image quality, the performance hit will be higher though. In general, increasing the levels of anti-aliasing and anisotropic filtering will reduce some FPS and you will have to decide these by trial and error.



Anisotropic filtering results in better texture details

Catalyst A.I. and Mipmap

Catalyst A.I. section allows you to decide the level of optimisations for certain applications. There exist certain shortcuts that enable the ATI graphics card to perform better with almost no loss in quality in certain games and benchmarks, this option allows the Catalyst to recognise these programs and automatically modify the driver settings for better performance while that application is being run. You can control the level of optimisation by moving the slider. Crossfire users should move this slider to Advanced (to the right) to force Alternate Frame Rendering mode in Direct3D games. Disable Catalyst A.I. only for troubleshooting purposes. Next is the Mipmap Detail Level, where you can move the slider to determine the level of sharpness of image. It is best to keep this slider near the middle, because you will see that the FPS will quickly deteriorate as you go higher. The All Settings sub-section contains all the settings that we have adjusted, but in a single page.

Direct3D And OpenGL

You may have noticed that there are no separate sections for Direct3D and OpenGL settings in CCC. This is taken care of in More Settings. Under Direct3D, you will find Enable geometry instancing. This should be enabled to optimise the rendering of 3D objects that appear multiple times in a 3D scene by uploading one instance of the object and then replicating it for as many times as required. Rendering of grass, leaves and ripples are some of the classic examples taking use of this feature. Support DXT texture formats